7.3 Slope Fields

Name _____ Period ____ Date _____

AP Calculus 7.3 · Student sheet

I can…

- find the slope of the segment at a point in a slope field.
- match a slope field to its differential equation.
- sketch a solution curve that follows a slope field.

Vocabulary · fill in the missing word

Slope field — short _____ at grid points with slope $\frac{dy}{dx} = f(x, y)$

Solution curve — the graph of a _____ solution; it is tangent to the segments it passes through

Part 1 · Build a slope field

dy/dx = x − y

y \ x	−1	0	1
1			
0			
−1			

Part 2 · Notes and examples (fill in each note, then work the example)

Slope field: at each point (x, y), draw a short segment with slope _____.

Example 1

For $\frac{dy}{dx} = x + y$, find the slope at (1, 2), (−1, 1), and (0, 0).

Horizontal segments: where dy/dx = _____.

Example 2

Where are the segments horizontal for $\frac{dy}{dx} = x - y$?

Matching: test a few _____ and compare their slopes.

Example 3

Match a slope field to its equation by testing points.

Solution curves are _____ to the segments they pass through.

Example 4

In the slope field for $\frac{dy}{dx} = y$, sketch the solution through (0, 1).

Part 3 · Partner practice

Partner 1

For $\frac{dy}{dx} = xy$, find the slope at $(2, -1)$.

Partner 2

(AP® Practice) For the slope field of $\frac{dy}{dx} = x^2$, which is true?

(A) The slopes are the same along each vertical line and never negative.

(B) The slopes are the same along each horizontal line.

(C) The slopes are negative when $x < 0$.

(D) Every segment is horizontal.